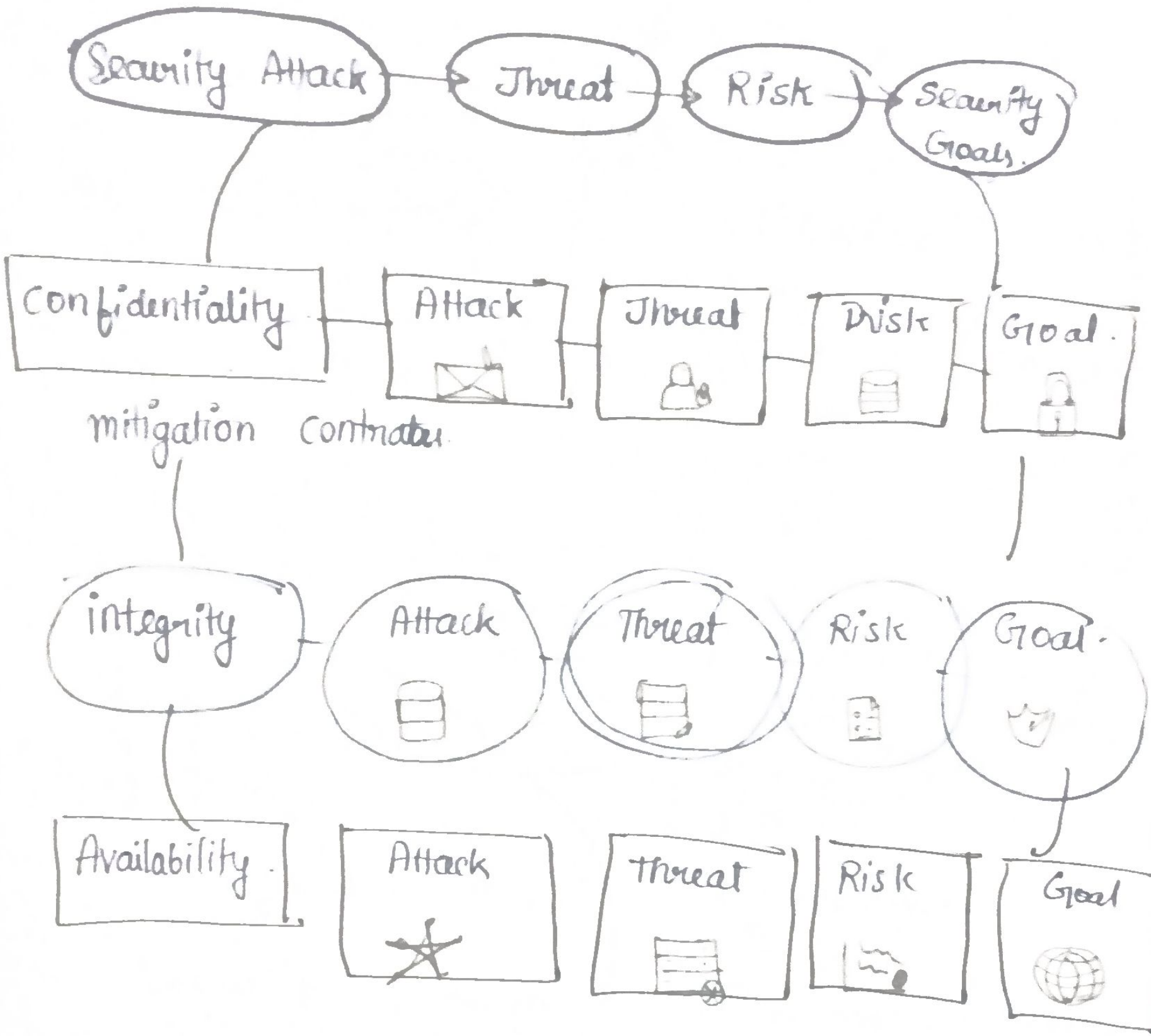


Attack → Action

Threat → potential cause of Harm

Risk → Likelihood x impact

CIA goals → confidentiality

Intregrity

Availability

Substitution Techniques.

A B C ... Y
| | | | |
Q D M L

Monalphabetic Cipher

Keytype
Strength
Complexity

A B C
| | |
X Y Z. Ciphers

Caesar cipher

Key Type
Strength
Complexity

[a b c
d e f
g h i] mod 26

Hill cipher.

Key Type
Strength
Complexity

K E Y
| | |
O O O.

Vigenere Cipher.

Keytype
Strength.
Complexity

Classical Encryption Schemes.

Substitution Techniques
(Replace Symbols)

- Letters are Substituted with Each other Letters.
- order of Letter remains the Same
- Data Structure Effect
- Security level

Transposition Techniques.
(rearrange position)

- Letters are rearranged in a different order.
- Actual Letters remains the Same
- Data Structure
- Security Level

Steganography

Data Hiding

- Hide data in image, audio, video

Watermarking

- Embedding information into digital

Survivability

- Ensures the detectable after procesing

Contributes to Secure Communication

How each contributes

Real world Application

Relationship